A finite-history Lyapunov function for geodesic inertial proximal trajectories

Abstract

We study an inertial proximal recurrence on a Hadamard manifold of finite dimension and globally lower-bounded sectional curvature. For a proper lower semicontinuous geodesically convex objective with an attained minimum, every constant inertia in $[0, 1)$ and every positive proximal parameter admit an explicit Lyapunov function of the current two-point phase and objective value. Its coercivity and one-step descent bound every trajectory from arbitrary initial points, without coercivity of the objective or boundedness of its minimizer set. The same energy proves square summability of the increments and convergence to one minimizer. A regularized distance with globally bounded Hessian supplies the geometric term, and no future trajectory terms are used. Classical ingredients, Euclidean ancestry and unresolved comparisons with nearby literature are identified.

Keywords: Lyapunov function; inertial proximal dynamics; Hadamard manifold; boundedness; trajectory convergence

2020 Mathematics Subject Classification. Primary 65K10; Secondary 53C21, 37B25.

1 Introduction

A decreasing objective–kinetic energy can control the increments of an inertial trajectory without controlling its positions. On a curved space, a distance term must also accommodate the error produced by geodesic extrapolation. The purpose of this note is to give a finite-history Lyapunov function which performs both tasks for one precise proximal recurrence.

The Euclidean convex inertial proximal method and its constant-inertia range $0 \leq \alpha < 1$ are already covered by Alvarez [1]. Proximal methods on Riemannian manifolds were studied by Ferreira and Oliveira [2]. The squared-distance proximal comparison and the pointwise Hessian comparison used below are standard. Our geometric estimate uses the latter in the form recalled in Appendix B of Alimisis et al. [3]. The construction combines those ingredients with a regularized distance and an explicit coefficient balance.

We keep the global lower-curvature assumption throughout the argument. In particular, an orbit bound is a conclusion of the energy estimate, rather than a premise used to choose a curvature constant. The result concerns the literal recurrence below with constant inertia and an arbitrary original two-point initialization. It does not address general equilibrium resolvents, over-relaxed updates or variable inertial parameters. Section 5 and the accompanying verification note record the limits of the source comparison; no complete priority claim is made.

2 The recurrence and the Lyapunov theorem

Let M be a finite-dimensional Hadamard manifold of positive dimension, with Riemannian metric g, distance d, and global sectional-curvature bounds

$$-\kappa^2 \le \mathrm{Sec}_M \le 0, \qquad 0 \le \kappa < \infty. \tag{1}$$

Thus M is complete and simply connected, and its exponential map at every point has a global inverse, denoted by log. Let $f : M \to \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous and geodesically convex. Suppose

$$S = \operatorname*{argmin}_M f \neq \varnothing, \qquad f_* = \min_M f \in \mathbb{R}. \tag{2}$$

Fix $\lambda > 0$ and $0 \le \alpha < 1$. Starting from any $x_0, x_1 \in M$, consider

$$\begin{aligned} y_n &= \exp_{x_n}(-\alpha \log_{x_n} x_{n-1}), \\ x_{n+1} &= \operatorname*{argmin}_{u \in M} \left\{ f(u) + \frac{d(u, y_n)^2}{2\lambda} \right\}, \qquad n \ge 1. \end{aligned} \tag{3}$$

The sign in the first line extrapolates beyond x_n along the geodesic arriving from x_{n-1}. The initial points need not belong to $\operatorname{dom} f$ or coincide.

Write $d_n = d(x_n, x_{n-1})$. For any fixed $z \in S$, define

$$\phi_z(x) = \sqrt{1 + d(x,z)^2}, \quad \phi_n = \phi_z(x_n), \quad c = \frac{(1+\kappa)\alpha(1+\alpha)}{2}, \quad \mu = 1 + \frac{c}{1-\alpha}. \tag{4}$$

For the original indices $n \ge 2$, set

$$E_n = \lambda\big(f(x_n) - f_*\big) + \frac{\alpha}{2} d_n^2, \tag{5}$$

$$H_n(z) = \phi_n - \alpha\phi_{n-1} + c d_n^2 + \mu E_n + \alpha. \tag{6}$$

These are functions of finite current states. The chosen z and f_* are analysis references; the recurrence does not need to know them or evaluate the energies.

Theorem 1. *Under* (1)–(2)*, the recurrence* (3) *is uniquely defined for every initial pair. Its objective values are finite from x_2 onward. For every $z \in S$ and every $n \ge 2$, the energies* (5)–(6) *satisfy*

$$E_{n+1} - E_n \le -(1-\alpha) d_{n+1}^2 - \frac{\alpha}{2}(d_{n+1} - d_n)^2, \tag{7}$$

$$H_n(z) \ge (1-\alpha)\phi_n + \left(c + \frac{\alpha}{4}\right) d_n^2 + \mu\lambda\big(f(x_n) - f_*\big), \tag{8}$$

$$H_{n+1}(z) - H_n(z) \le -(1-\alpha) d_{n+1}^2 - \frac{\mu\alpha}{2}(d_{n+1} - d_n)^2. \tag{9}$$

In particular,

$$d(x_n, z) \le \phi_n \le \frac{H_2(z)}{1-\alpha} \quad (n \ge 2), \qquad \sum_{j=3}^{\infty} d_j^2 \le \frac{E_2}{1-\alpha}. \tag{10}$$

There exists $x_\infty \in S$ such that

$$x_n \longrightarrow x_\infty, \qquad y_n \longrightarrow x_\infty, \qquad f(x_n) \longrightarrow f_*, \qquad d(x_{n+1}, y_n) \longrightarrow 0. \tag{11}$$

All convergence of points is convergence in the metric d.

For $\alpha > 0$, (8) bounds both coordinates of every sublevel of the phase function in (6). At $\alpha = 0$, the map is first-order and the energy does not constrain an arbitrary unused predecessor. Its bound on consecutive actual positions still bounds the original phase. A radius covering every original position is

$$R_z = \max\left\{ d(x_0, z), d(x_1, z), \frac{H_2(z)}{1-\alpha} \right\}. \tag{12}$$

The differences $d_{n+1} - d_n$ in the theorem are differences of scalar speeds. They are not differences of tangent vectors at distinct base points. Neither finite trajectory length nor selection of a distinguished minimizer is asserted.

3 Proximal dissipation and regularized distance

We first establish the two estimates entering the energy. The first is purely proximal and gives square-summable increments before any position bound is known.

Lemma 1. *For every $y \in M$, the function $u \mapsto f(u) + d(u, y)^2/(2\lambda)$ has a unique minimizer p, with $f(p) < \infty$. For every $w \in \operatorname{dom} f$,*

$$2\lambda\big(f(p) - f(w)\big) \le d(w, y)^2 - d(p, y)^2 - d(p, w)^2. \tag{13}$$

Consequently, the original iterates satisfy (7) for every $n \ge 2$.

Proof. Since $f \ge f_*$, each finite sublevel of the proximal objective is bounded. It is closed by lower semicontinuity and compact by the finite-dimensional Hopf–Rinow theorem. A finite comparison value is supplied by any point of S, so a minimizing sequence has a convergent subsequence and the minimum is attained.

For the constant-speed segment γ from p to w, the Hadamard squared-distance inequality gives

$$d(\gamma(t), y)^2 \le (1-t)d(p, y)^2 + td(w, y)^2 - t(1-t)d(p, w)^2, \quad 0 \le t \le 1. \tag{14}$$

Combining this with geodesic convexity of f shows that the proximal objective is strongly convex and has a unique minimizer. Comparing its value at p with its value at $\gamma(t)$, dividing by $t > 0$ and letting $t \downarrow 0$ proves (13).

The extrapolation in (3) is globally defined and satisfies

$$d(x_n, y_n) = \alpha d_n. \tag{15}$$

Induction therefore defines all the iterates, with $x_n \in \operatorname{dom} f$ for every $n \geq 2$. In (13), put $p = x_{n+1}$, $y = y_n$ and $w = x_n$, using only $n \geq 2$. The reverse triangle inequality gives

$$d(x_{n+1}, y_n) \geq |d_{n+1} - \alpha d_n|.$$

It follows that

$$\lambda\big(f(x_{n+1}) - f(x_n)\big) \leq -d_{n+1}^2 + \alpha d_n d_{n+1}. \tag{16}$$

Adding $\alpha(d_{n+1}^2 - d_n^2)/2$ proves (7). In particular, $E_n \geq 0$ has a finite limit, and finite telescoping gives

$$(1-\alpha)\sum_{j=3}^{N+1} d_j^2 \leq E_2 - E_{N+1} \leq E_2, \qquad N \geq 2. \tag{17}$$

Thus $\sum_{n=2}^{\infty} d_n^2 < \infty$ and $d_n \to 0$. This argument does not use boundedness of the positions or finite values of $f(x_0)$ and $f(x_1)$. ∎

The next lemma replaces squared distance by a smooth radial function whose Hessian has a global upper bound. This is where the lower-curvature hypothesis enters.

Lemma 2. *For every $z \in M$, the function $\phi_z(x) = \sqrt{1 + d(x,z)^2}$ is smooth and satisfies*

$$\|\operatorname{grad} \phi_z\| \leq 1, \qquad 0 \leq \operatorname{Hess} \phi_z \leq (1+\kappa)g. \tag{18}$$

For all $x, u \in M$,

$$\phi_z\big(\exp_x(-\alpha \log_x u)\big) \leq (1+\alpha)\phi_z(x) - \alpha\phi_z(u) + c\, d(x,u)^2. \tag{19}$$

Consequently, every trajectory in Theorem 1 satisfies

$$\phi_{n+1} - (1+\alpha)\phi_n + \alpha\phi_{n-1} \leq c d_n^2, \qquad n \geq 1. \tag{20}$$

Proof. Away from z, put $r = d(\cdot, z)$. The pointwise Hessian comparison under (1) gives, for v perpendicular to $\operatorname{grad} r$,

$$0 \leq \operatorname{Hess} r(v,v) \leq h_\kappa(r)\|v\|^2, \qquad h_\kappa(r) = \begin{cases} \kappa \coth(\kappa r), & \kappa > 0, \\ 1/r, & \kappa = 0. \end{cases} \tag{21}$$

The radial and mixed components of $\operatorname{Hess} r$ vanish. These are the standard pointwise comparison factors recalled in [3, Appendix B]. We use them along each finite radial geodesic under the given global curvature bound, without imposing a diameter bound on an unknown orbit.

Direct differentiation yields

$$\operatorname{grad} \phi_z = \frac{r}{\sqrt{1+r^2}} \operatorname{grad} r, \qquad \operatorname{Hess} \phi_z = \frac{dr \otimes dr}{(1+r^2)^{3/2}} + \frac{r}{\sqrt{1+r^2}} \operatorname{Hess} r. \tag{22}$$

For $t > 0$, the inequality $e^{2t} - 1 \geq 2t$ implies

$$t \coth t = t + \frac{2t}{e^{2t}-1} \leq t + 1.$$

When $\kappa > 0$, the transverse upper coefficient in (22) is therefore at most

$$\frac{1+\kappa r}{\sqrt{1+r^2}} \le 1+\kappa.$$

When $\kappa = 0$, that coefficient is at most one. The radial coefficient is also at most one, and all these coefficients are nonnegative. At z, normal coordinates give $\phi_z(\exp_z v) = \sqrt{1+\|v\|^2}$, so the function is smooth and its Hessian is g_z. This proves (18) everywhere. It also proves the global Lipschitz estimate

$$|\phi_z(x) - \phi_z(u)| \le d(x,u). \tag{23}$$

Consider the complete geodesic $\gamma(t) = \exp_x(t \log_x u)$ on $[-\alpha, 1]$, of constant speed $D = d(x,u)$. If $b(t) = \phi_z(\gamma(t))$, then $b''(t) \le (1+\kappa)D^2$. Hence

$$t \longmapsto b(t) - \frac{(1+\kappa)D^2}{2}t^2$$

is concave. Applying its concavity at 0, between $-\alpha$ and 1, gives (19). The argument includes $D = 0$ and $\alpha = 0$; in the latter case the estimate is equality.

Finally, (13) with $w = z \in S$ implies

$$0 \le 2\lambda\big(f(x_{n+1}) - f_*\big) \le d(y_n,z)^2 - d(x_{n+1},y_n)^2 - d(x_{n+1},z)^2. \tag{24}$$

Thus $d(x_{n+1},z) \le d(y_n,z)$ and $\phi_z(x_{n+1}) \le \phi_z(y_n)$. Combining this with (19) for $(x,u) = (x_n, x_{n-1})$ proves (20). ∎

4 Capture and convergence of the actual trajectory

Proof of Theorem 1. Lemma 1 proves well-posedness, finite objective values from the original index two, (7) and square summability. We now use Lemma 2 to control the positions.

Fix $z \in S$. From (23), $\phi_{n-1} \le \phi_n + d_n$. Substitution in (6) and completion of a square give

$$\begin{aligned} H_n(z) \ge (1-\alpha)\phi_n + \Big(c + \frac{\alpha}{4}\Big)\, d_n^2 + \mu\lambda\big(f(x_n) - f_*\big) \\ + \frac{\alpha}{4}(d_n - 2)^2 + \frac{\alpha(\mu-1)}{2} d_n^2. \end{aligned} \tag{25}$$

Since $\mu \ge 1$, this proves (8) and $H_n(z) \ge 1 - \alpha > 0$. For $\alpha > 0$, the same calculation for an arbitrary pair (u,x) with $f(x) < \infty$ shows that a sublevel bounds $d(x,z)$ and $d(x,u)$, hence also $d(u,z)$. This is coercivity in both phase coordinates. No bound on f-sublevels or on S was used.

For the one-step difference, (20) and (7) yield

$$\begin{aligned} H_{n+1}(z) - H_n(z) &\le cd_n^2 + c(d_{n+1}^2 - d_n^2) + \mu(E_{n+1} - E_n) \\ &\le \big(c - \mu(1-\alpha)\big)d_{n+1}^2 - \frac{\mu\alpha}{2}(d_{n+1} - d_n)^2. \end{aligned}$$

The identity $\mu(1-\alpha) = 1-\alpha+c$ gives (9). The energy is bounded below and nonincreasing, so it has a finite limit. Its lower bound and descent prove (10); including the original finite pair gives (12). At $\alpha = 0$, $H_n = \phi_n + \lambda(f(x_n) - f_*)$, and the same bound applies to every actual position. This also bounds the actual consecutive phases at that endpoint. Capture has now been established without compactness or a future trajectory term.

We next prove that the distance to every fixed point of S has a limit. Both $H_n(z)$ and E_n have finite limits, while $d_n \to 0$. By (6), the quantity

$$A_n(z) = \phi_n - \alpha\phi_{n-1} = H_n(z) - cd_n^2 - \mu E_n - \alpha$$

has a finite limit. Moreover, (23) gives $|\phi_n - \phi_{n-1}| \le d_n \to 0$. The identity

$$A_n(z) = (1-\alpha)\phi_n + \alpha(\phi_n - \phi_{n-1})$$

then shows that ϕ_n converges. Since $\phi_n \ge 1$, the distance $d(x_n, z) = \sqrt{\phi_n^2 - 1}$ also has a limit. This conclusion holds for every fixed $z \in S$ before any cluster point has been chosen.

The proximal displacement satisfies

$$\rho_n := d(x_{n+1}, y_n) \le d_{n+1} + \alpha d_n \longrightarrow 0. \tag{26}$$

Fix $z \in S$ again and put $r_n = d(x_{n+1}, z)$. The triangle inequality gives $d(y_n, z) \le r_n + \rho_n$. Using (24), we obtain

$$0 \le \lambda\big(f(x_{n+1}) - f_*\big) \le \frac{(r_n + \rho_n)^2 - \rho_n^2 - r_n^2}{2} = r_n\rho_n \longrightarrow 0, \tag{27}$$

since r_n is bounded by the already proved capture estimate.

The orbit lies in the closed ball from (12), which is compact by Hopf–Rinow. Choose a convergent subsequence $x_{n_j} \to x_\infty$. Lower semicontinuity and (27) imply $f(x_\infty) \le f_*$, so $x_\infty \in S$. The distance $d(x_n, x_\infty)$ has a limit by the preceding argument, and that limit is zero along the chosen subsequence. Hence the whole trajectory converges to x_∞. Finally, $d(y_n, x_n) = \alpha d_n \to 0$ proves convergence of y_n to the same point. Equations (26) and (27) give the remaining conclusions. ∎

5 Scope and relation to earlier results

The role of the geometric term is specific. The kinetic energy E_n gives square-summable increments, but this alone does not bound the total displacement. The regularized distance ϕ_z has a globally bounded Hessian under (1). Its extrapolation error is

therefore a fixed multiple of d_n^2, and that error is absorbed in the one-step descent of H_n. The extra constant α and the square completion in (25) give a direct bound on the actual phase. Compactness is used only after this bound has been proved.

Neither the pointwise Hessian comparison nor the Euclidean inertial parameter range is claimed as new. In flat space, (3) becomes

$$x_{n+1} = \operatorname{prox}_{\lambda f}\big(x_n + \alpha(x_n - x_{n-1})\big),$$

whose convex constant-inertia convergence is classical [1]. The noninertial Riemannian proximal method is also classical [2]. Our statement is a self-contained energy and trajectory result for the nonflat class in (1)–(2).

The lower-curvature hypothesis is global and fixed in advance. It is not replaced by the assertion that a bounded trajectory would encounter only a bounded curvature range. The proof of Lemma 2 uses the pointwise comparison recalled in [3, Appendix B], rather than applying a bounded-diameter formulation to an as-yet unbounded trajectory. It does not require an upper extrapolation identity for the bare squared distance.

Several nearby inertial proximal and inertial Mann papers on Hadamard manifolds have only partially accessible primary texts. In particular, the exact assumptions of the papers identified by DOIs 10.1080/00036811.2021.2016719, 10.1007/s12597-024-00838-1 and 10.4208/nmtma.OA-2023-0039 have not been fully compared here. These are material limits on any assertion of priority. The accompanying verification and source note distinguishes inspected update formulas from unresolved theorem coverage. Theorem 1 makes no claim about failure of any broader algorithm in that literature.

Finally, the conclusion is convergence to some minimizer for the specified initialized orbit. It does not identify that minimizer in advance, provide a convergence rate, or imply $\sum_n d_n < \infty$. The bounds deteriorate as $\alpha \uparrow 1$, and the endpoint $\alpha = 1$ is outside the statement. Infinite-dimensional spaces, variable inertia, over-relaxation and spaces without a given global lower-curvature bound are not included.

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